

Classification of PCA based Reduced Deep Features by SVM for Diagnosing Lung and Colon Cancer

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Abstract—Medicine has developed in recent years at all levels, especially in diagnosing cancer, which is one of the most dangerous and deadly diseases. However, it is still the main cause of death for the largest percentage of people. Particularly, colon and lung cancer constitutes 20% of all cancer cases. The early diagnosis of the cancer stages increases the prognosis and likelihood of successful treatment. In this regard, artificial intelligence based assistance toolboxes play critical roles in diagnosing stages. In this study, three lightweight deep learning networks were utilized in the classification of colon and lung cancer images under three scenario. AlexNet, SqueezeNet, and ShuffleNet were employed as pre-trained models in scope of transfer learning idea. In the first scenario, CNNs were utilized as a classifier for lung and colon cancer images. The classification performances of each model were reported as SoftMax performances, individually. In the second scenario, only the features extracted from the images by the pre-trained model were collected and fed to the SVM classifier. In the last scenario, a PCA was applied over the extracted features of the networks to reduce the number of features. After the feature selection process, the reduced feature sets were also fed to SVM to raise the classification accuracy. The highest results were achieved using the third scenario for both colon and lung cancer images. Especially using the ShuffleNet, the classification was measured as 99.93% and 97.92% for colon and lung cancer datasets, respectively.

Index Terms—Deep Features, SoftMax Classification, SVM, PCA, Lung Cancer, Colon Cancer

I. INTRODUCTION

According to the reports of the World Health Organization (WHO) and many studies, cancer is considered as one of the most deadly diseases in humans because of its high spreading speeds [1]. Different cancer types have been diagnosed so far. Lung cancer represents the highest percentage of risk, as it constitutes about 11.6% of the incidence of cancers and a rate of about 18.4% of all cancer origin deaths. Similarly, Colon cancer has an effect of 9.2% on cancer deaths [2]. There are still significant increases in those percentages due to increase the number of populations and smokers, unhealthy foods, air pollution etc. [3].

Cancer cells are usually represented by foreign cells that arise in the organ that they infect and spread very quickly. Recent reports by the World Health Organization found that cancer affects poor areas at a greater rate. The reason may be attributed to a lack of awareness and knowledge of early examinations of diseases [4]. However, early diagnosis and treatment at the beginning of the disease before it spreads is the best way to protect against this deadly disease that daily leads to the lives of many people. Despite this, there are millions of people in different parts of the world who lack awareness and early diagnosis methods to prevent the disease [5]. To solve this problem, machine learning techniques which is a part of artificial intelligence was utilized to diagnose many diseases based on the patient's symptoms. According to many experiments and measurements, an advanced sub-category of machine learning, Deep Learning (DL), overcomes many other modern methods in the field of classification and prediction for medical data. The core of DL is the Convolutional Neural Networks (CNNs) which have started to become mainstream for various fields such as in medicine since 2014 [6]. In the classification of cancer diseases, DL based algorithms have been also implemented to analyze the histopathological images for saving the personnel time and effort while meeting the requirements of high accuracy [7] [8].

In this study, three deep learning based network architectures were utilized in the classification of the histopathology images including two cancer types. Three classes of lung cancer and two classes of colon cancer were aim to classify by three pre-trained lightweight networks (AlexNet, SqueezeNet, ShuffleNet). Model performance analysis have been reported for colon and lung datasets individually in three scenario. In the first scenario, each model was used as a deep learning classifier over datasets. The classification performances of the softmax layers are given. The second scenario was performed to report the feature extraction performance of each networks. Support Vector Machines (SVM) was used as classifier. In the last scenario, Principal Component Analysis (PCA) was implemented to reduce the dimensionality of the CNNs

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features by creating new features containing most of the original data variance followed by SVM [9].

II. RELATED WORK

Bijaya Kumar Hatuwal used the lung cancer images of LC25000 dataset with three classes (benign tissue, Adenocarcinoma and squamous carcinoma cells) and 5000 histopathology images for each class. They designed a custom network including three hidden layers, one input layer, and one fully connected layer. The dataset were split by a ratio of 90:10 for training and validation. They trained the model for 20 epochs with batch size 64. Their model achieved an accuracy of 0.9611 for training and 0.9720 for validation for the final epoch [10].

Similarly, Mohammad Ali Abbas et al. used just lung in LC25000 dataset to determine the effectiveness of convolutional neural networks. Six types of convolutional neural networks were used in this work. They achieved 0.973, 0.997, 0.986, 0.992, 0.999, and 0.999 accuracy scores for VGG-19, AlexNet, ResNet-18, ResNet-34, ResNet-50, and ResNet-101, respectively [11].

Tasnim et al. used the MobileNetV2 deep learning architecture to classify the colon cancer tissues under two categories as Colon adenocarcinoma (cancerous) and Colon benign tissue (not cancerous). They compared two methods of pooling for MobileNetV2 applications, the first is max pooling and the second is average pooling. The result of max pooling experiments was measured as 0.9749, while the average pooling resulted in 0.9548 accuracies [12].

Missouri et al. used the benchmark datasets obtained from the LUNA16 challenge. Datasets include 888 CT scans in MetaImage format (mhd/raw) filtered out from the public Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI). Pre-trained network was used to extract the image features, then combined features were fed to SVM, AdaBoost, k-Nearest Neighbor (KNN), random forest, and fully connected layer (FC) to classify nodules and non-nodules. VGG16 and VGG19 gave better results when combined with SVM to detect lung nodules with an accuracy of 0.9199 [13].

Zhenghao Shi et al. performed their methods over dataset consisting of 1018 cases from seven institutions, and each case consists of images from a thoracic CT scan. In this work, pre-trained VGG16 networks were used to extract the important features in the image, and then the features obtained from the previous step were introduced into the SVM to determine whether the filter entered is a node or not and the results of 0.878 and a sensitivity of 0.872 were obtained [14].

III. MATERIALS AND METHODS

A. Dataset Information

In this study, the LC25000 data set was used for colon and lung cancer images, which were obtained from the Kaggle website [15]. It is a large, clear, and good-quality data set consisting of 25000 medical images with five categories of

lung and colon cancer. All images are 768 x 768 pixels in jpeg format. This dataset is from an original sample from HIPAA-compliant and validated sources containing 500 images of colon tissue (250 benign colon tissue (BN) and 250 colon adenocarcinomas (ACA)) and 750 images of lung tissue (250 benign lung tissue (BN), 250 lung adenocarcinomas (ACA), and 250 lung squamous cell carcinoma (SCC)). The number of images was increased to 25,000 using the condenser package, contains the same categories, each category has a size of 5000 images. Example images for each class are demonstrated in Figure 1. Tests were performed separately for the colon and lung sub-sets.

B. Transfer learning

CNN is a fully connected neural network that is preceded by several layers of convolutional filters. In the transfer learning methodology (TL), CNNs that have been previously pre-trained by large and varied amount of data were used in a new domain. The previously acquired features are transferred and customized. Transfer learning saves time in the model training, instead of starting to train the model from scratch. TL is performed using the pre-trained weights as a starting point for the new training process. In this presented study, three lightweight CNNs, namely AlexNet, SqueezeNet, ShuffleNet, were utilized to classify colon and lung cancer datasets under the transfer learning idea. Three scenario have been tested. In the first scenario, the pre-trained model classification performances were measured individually. Then, pre-trained networks were employed as feature extractor for the SVM classifier. The feature extraction performances of pre-trained lightweight networks were reported. Lastly, a feature selection algorithm, Principle Component Analysis, has been performed over the extracted features of pre-trained models to observe the feature set organization effect on the classification performances.

1) *AlexNet*: Despite the continuous development in the field of transfer learning and the emergence of many convolutional neural networks, AlexNet, which was proposed in 2012, is still considered one of the pioneering and reliable networks for the image classification problems. Alexnet's structure consists of 19 layers including five two-dimensional convolutional layers, followed by three fully connected layers (FC), and at the end of it after extracting the features, three fully connected layers (FC) [16]

2) *SqueezeNet*: SqueezeNet is considered one of the smallest convolutional neural networks, as its 50 times smaller than Alexnet, yet it gave the same classification power and was three times faster than Alexnet. It is widely used in the medical field for its speed and power of implementation. The structure of SqueezeNet consists of an independent convolutional layer, which is responsible for receiving the input images, followed by seven fire units, and ending with a convolutional layer. SqueezeNet

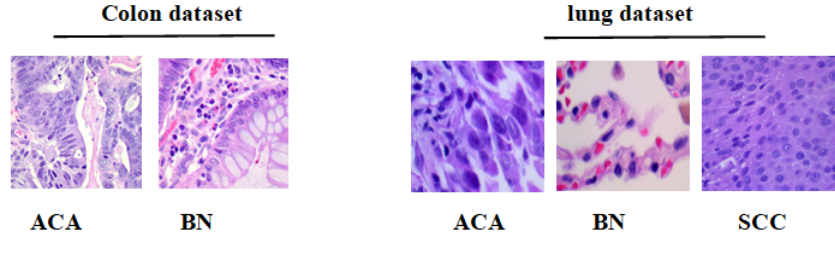


Fig. 1. Example images for each class located in LC25000 dataset

is characterized by being composed of a series of pressure represented by 1×1 of filters and expansion represented by a series of layers 1×1 and 3×3 , and the joint of both layers is called the fire unit [17].

3) *ShuffleNet*: The network structure consists of 50 layers. This network features two new processes. The first process is group convolution, and the second process is a channel switch point. The benefit of this process is to reduce the complexity. Point group convolutions to reduce the computing complexity of the $1 \times$ convolution and also to operate the channels for the transfer of distinguished information through the channels. It consists of three random unit variables [18].

C. Principal Components Analysis (PCA)

In the feature selection stage, Principal Components Analysis (PCA) was used, which will reduce the dimensionality of the features by creating new features containing most of the original data variance. PCA extracts important data from the image without affecting the original image data as similar properties are combined with each other. PCA significantly reduces the dimensions of the space without changing the number of original variables, as it uses all of the original variables to generate new latent variables (principal components) [19]

D. Support Vector Machine (SVM)

It is one of the supervised machine learning algorithms that give good results in the field of image classification within more short time than other complex techniques. The SVM algorithm's work depends on finding the best decision line or boundary. This limit or the best region is called the super level. The work of SVM is focused on finding the closest point to the dividing line for all categories. In this work, in the case of more than two categories. The curve is used to separate the three categories as in the lung part. These points are called support vectors, and the distance between vectors and the super level is called the margin. SVM is an excellent technique for both linear and non-linear relations modelling. In this work linear kernel was used [9]

E. Performance Metrics

Accuracy, sensitivity, and recall scores —evaluation measures based on confusion matrices— were used in

the studies. In order to create such measurements, four values—True Positive (TP), False Positive (FN), True Negative (TN), and False Negative (FN)—were collected from the confusion matrix of test results. True positives demonstrate that the observation and the successful forecast are the same. A false positive occurs when the observation is negative but the anticipation is positive. True negative shows how accurately negative sample findings may be identified. Another example of incorrect labeling samples as negative, when they fact contains positive labels, is false negative. The projections in TN and TP are precise. However, FN and FP demonstrate how the models' incorrect predictions. The tests utilized precision, sensitivity, and The accuracy score, which is regarded as the most important performance statistic once the data are balanced, represents the percentage of positive and negative samples that were correctly classified in the data set. The accuracy value is insufficiently revealing for some classification problems with inconsistent findings and unbalanced data, though. These problems should include recall and precision scores. The harmonic mean of the precision and recall metrics, known as the F1-score, is also more enlightening for the comparison. By using the harmonic mean, outliers can have less of an impact. The formula for the performance metrics is given in the following equations.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

F. Methodology

In this work, three scenario were followed to classify a group of colon and lung cancer images. Three pre-trained networks as Alexnet, SqueezeNet, and Shufflenet were considered as the CNNs. In the first scenario, CNNs were utilized as a classifier for lung and colon cancer images. The classification performances of each model were reported as SoftMax performances, individually. In the second scenario, only the features extracted from the images

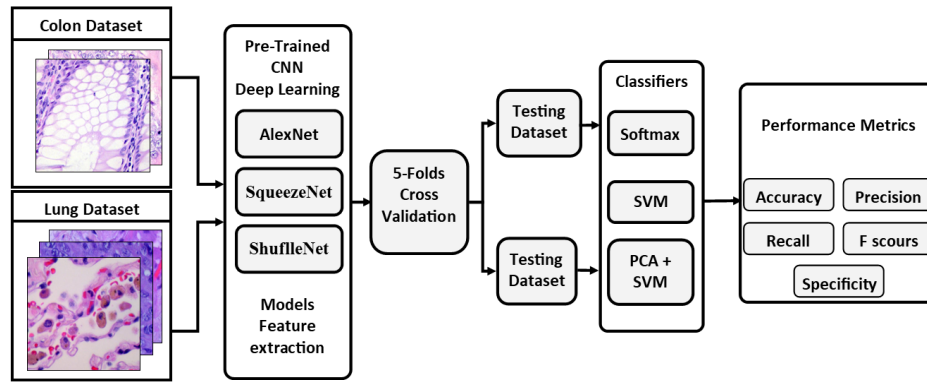


Fig. 2. Flowchart of the Presented Study

by the pre-trained model were collected and fed to the SVM classifier. In the last scenario, a PCA was applied over the extracted features of the networks to reduce the number of features, especially the unnecessary ones. After the feature selection process, the reduced feature sets were also fed to SVM to raise the classification accuracy. The highest results were obtained by using the third scenario for both colon and lung cancer images. The general flowchart of the study is demonstrated in Figure 2.

IV. EXPERIMENTAL RESULTS

All experiments were performed using Python Keras library over Google Collab environment. The classification performances of the models on the colon cancer dataset, which contains two classes (Adenocarcinoma, Benign), under three scenario are presented in Table I with the accuracy scores. In the individual performance evaluation of pre-trained networks which refers to SoftMax classification results in Table I, AlexNet, SqueezeNet and ShuffleNet achieved 89.61%, 93.55% and 88.76% accuracy scores for the classification of colon cancer dataset, respectively. In the second experimental scenario, networks were utilized as the feature extractor. The features were then fed to the SVM classifier. The results of this strategy were resulted in better accuracies than the individual usage of models with 99.21%, 99.62% and 99.93% accuracy scores for AlexNet, SqueezeNet and ShuffleNet, respectively. In the last scenario, PCA was employed to eliminate uninformative features. The reduced feature sets were then fed to SVM similar to second scenario. The classification performance was measured as the highest scores in this scenario with an accuracy of 99.22%, 99.65% and 99.93%, respectively.

The performances of the models with the same experimental scenarios on the lung cancer dataset, which contains three classes (adenocarcinoma, squamous Cell Carcinoma, and Benign) are presented in Table II. The individual performance of three networks (Softmax based classification of AlexNet, SqueezeNet and ShuffleNet) were measured with an accuracy rate of 93.12%, 91.05% and 95.64% for the lung cancer dataset, respectively. In the

TABLE I
LC25000 COLON DATASET RESULTS

	SoftMax	SVM	SVM+PCA
AlexNet	89.61	99.21	99.22
SqueezeNet	93.55	99.62	99.65
ShuffleNet	88.76	99.93	99.93

TABLE II
LC25000 LUNG DATASET RESULTS

	SoftMax	SVM	SVM+PCA
AlexNet	93.12	97.43	97.44
SqueezeNet	91.05	96.87	96.92
ShuffleNet	95.64	97.91	97.92

second strategy, similar to the classification of colon cancer dataset, the results were also better than the first strategy with accuracy of 97.43%, 96.87% and 97.91%, respectively. In the last scenario, PCA provided a minimal increment to accuracies which achieved the best result as 97.44%, 96.92%, 97.92% for AlexNet, SqueezeNet and ShuffleNet, respectively. The confusion matrices of the best results achieved by ShuffleNet with SVM and PCA are given in Figure 3. In addition to accuracy, other performance metrics such as precision, recall, and f1-scores derived from the relevant confusion matrix are listed in Table III.

V. CONCLUSION

Cancer is generally considered among the deadliest diseases in people's lives, and in particular, colon and lung cancer constitutes a high percentage of the most dangerous cancers. Early detection of these tumors plays critical role on the treatment of disease. The implementation of artificial intelligence in medicine to classify medical images provides satisfactory results in analysis and eliminates the exhaustive time consuming processes. This paper also aims to determine a machine learning based analyzing approach to obtain the best and most accurate results for the classification of colon and lung cancer images. Three pre-trained CNNs namely AlexNet, SqueezeNet and ShuffleNet were employed under three scenario. In the

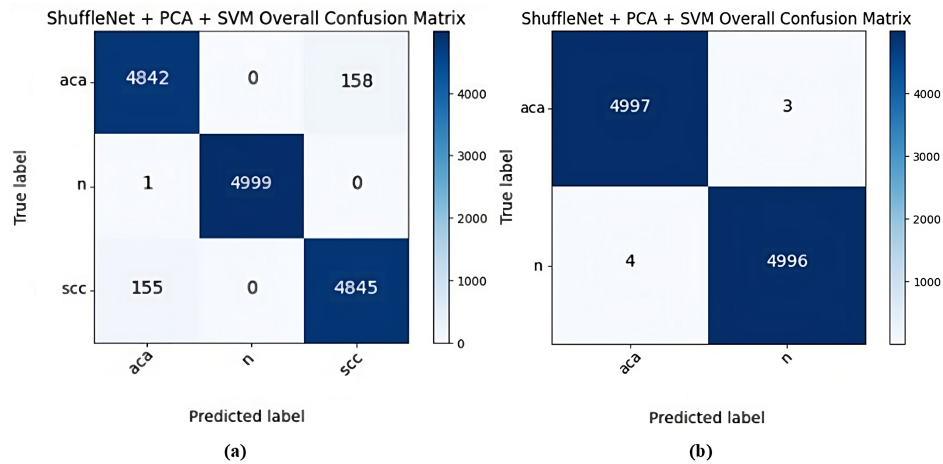


Fig. 3. Confusion Matrix of PCA and SVM Implementation on ShuffleNet Features (a) for Lung (b) for Colon

TABLE III
PERFORMANCE METRICS FOR PCA AND SVM IMPLEMENTATION ON SHUFFLENET FEATURES

	Lung			Colon	
	ACA	N	SCC	ACA	N
precision	0.97	0.1	0.97	1.0	1.0
Recall	0.97	0.1	0.97	1.0	1.0
F1 score	0.97	0.1	0.97	1.0	1.0
Accuracy	97.91%	99.99%	97.91%	99.94%	99.92%

first scenario, models were evaluated as classifiers by using softmax layers in the classification steps. In the second scenario, CNNs were used as feature extractors to feed a non-linear classifier SVM. Lastly, a feature selection technique, PCA, was applied over the extracted features to eliminate uninformative features. The last strategy using the PCA over the CNNs features to feed SVM resulted in the highest accuracies for both datasets.

REFERENCES

- [1] A. Bermúdez, I. Arranz-Salas, S. Mercado, J. A. López-Villodres, V. González, F. Rius, M. V. Ortega, C. Alba, I. Hierro, and D. Bermúdez, "Her2-positive and microsatellite instability status in gastric cancer—clinicopathological implications," *Diagnostics*, vol. 11, no. 6, p. 944, 2021.
- [2] C. Wu, M. Li, H. Meng, Y. Liu, W. Niu, Y. Zhou, R. Zhao, Y. Duan, Z. Zeng, X. Li *et al.*, "Analysis of status and countermeasures of cancer incidence and mortality in china," *Science China Life Sciences*, vol. 62, pp. 640–647, 2019.
- [3] T. P. Nguyen, H. N. Luu, M. V. T. Nguyen, M. T. Tran, T. T. V. Tuong, C. T. D. Tran, and P. Boffetta, "Attributable causes of cancer in vietnam," *JCO Global Oncology*, vol. 6, pp. 195–204, 2020.
- [4] B. White, M. Rafiq, A. Gonzalez-Izquierdo, W. Hamilton, S. Price, and G. Lyratzopoulos, "Risk of cancer following primary care presentation with fatigue: a population-based cohort study of a quarter of a million patients," *British Journal of Cancer*, vol. 126, no. 11, pp. 1627–1636, 2022.
- [5] L. F. Sanchez-Peralta, L. Bote-Curiel, A. Picon, F. M. Sanchez-Margallo, and J. B. Pagador, "Deep learning to find colorectal polyps in colonoscopy: A systematic literature review," *Artificial intelligence in medicine*, vol. 108, p. 101923, 2020.
- [6] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 2818–2826.
- [7] F. Zhuang, Z. Qi, K. Duan, D. Xi, Y. Zhu, H. Zhu, H. Xiong, and Q. He, "A comprehensive survey on transfer learning," *Proceedings of the IEEE*, vol. 109, no. 1, pp. 43–76, 2020.
- [8] X. Su, S. Xue, F. Liu, J. Wu, J. Yang, C. Zhou, W. Hu, C. Paris, S. Nepal, D. Jin *et al.*, "A comprehensive survey on community detection with deep learning," *IEEE Transactions on Neural Networks and Learning Systems*, 2022.
- [9] D. M. Abdullah and A. M. Abdulazeez, "Machine learning applications based on svm classification a review," *Qubahan Academic Journal*, vol. 1, no. 2, pp. 81–90, 2021.
- [10] B. K. Hatuwal and H. C. Thapa, "Lung cancer detection using convolutional neural network on histopathological images," *Int. J. Comput. Trends Technol*, vol. 68, no. 10, pp. 21–24, 2020.
- [11] M. A. Abbas, S. U. K. Bukhari, A. Syed, and S. S. H. Shah, "The histopathological diagnosis of adenocarcinoma & squamous cells carcinoma of lungs by artificial intelligence: A comparative study of convolutional neural networks," *MedRxiv*, pp. 2020–05, 2020.
- [12] Z. Tasnim, S. Chakraborty, F. Shamrat, A. N. Chowdhury, H. A. Nuha, A. Karim, S. B. Zahir, and M. M. Billah, "Deep learning predictive model for colon cancer patient using cnn-based classification," *Int. J. Adv. Comput. Sci. Appl*, vol. 12, pp. 687–696, 2021.
- [13] R. Mastouri, N. Khelifa, H. Neji, and S. Hantous-Zannad, "A bilinear convolutional neural network for lung nodules classification on ct images," *International Journal of Computer Assisted Radiology and Surgery*, vol. 16, pp. 91–101, 2021.
- [14] Z. Shi, H. Hao, M. Zhao, Y. Feng, L. He, Y. Wang, and K. Suzuki, "A deep cnn based transfer learning method for false positive reduction," *Multimedia Tools and Applications*, vol. 78, pp. 1017–1033, 2019.
- [15] A. A. Borkowski, M. M. Bui, L. B. Thomas, C. P. Wilson, L. A. DeLand, and S. M. Mastorides, "Lung and colon cancer histopathological image dataset (lc25000)," *arXiv preprint arXiv:1912.12142*, 2019.
- [16] S. Thalagala and C. Walgampaya, "Application of alexnet convolutional neural network architecture-based transfer learning for automated recognition of casting surface defects," in *2021 International Research Conference on Smart Computing and Systems Engineering (SCSE)*, vol. 4. IEEE, 2021, pp. 129–136.
- [17] L. S. Bernardo, R. Damaševičius, S. H. Ling, V. H. C. de Albuquerque, and J. M. R. Tavares, "Modified squeezeNet architecture for parkinson's disease detection based on keypress data," *Biomedicine*, vol. 10, no. 11, p. 2746, 2022.
- [18] N. Darapaneni, B. Krishnamurthy, and A. R. Paduri, "Convolution neural networks: A comparative study for image classification," in *2020 IEEE 15th International Conference on Industrial and Information Systems (ICIIS)*. IEEE, 2020, pp. 327–332.
- [19] M. Z. Nasution, "Face recognition based feature extraction using principal component analysis (pca)," *Journal of Informatics and Telecommunication Engineering*, vol. 3, no. 2, pp. 182–191, 2020.